



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO SRI PALANI MURUGAN PETROLEUM

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

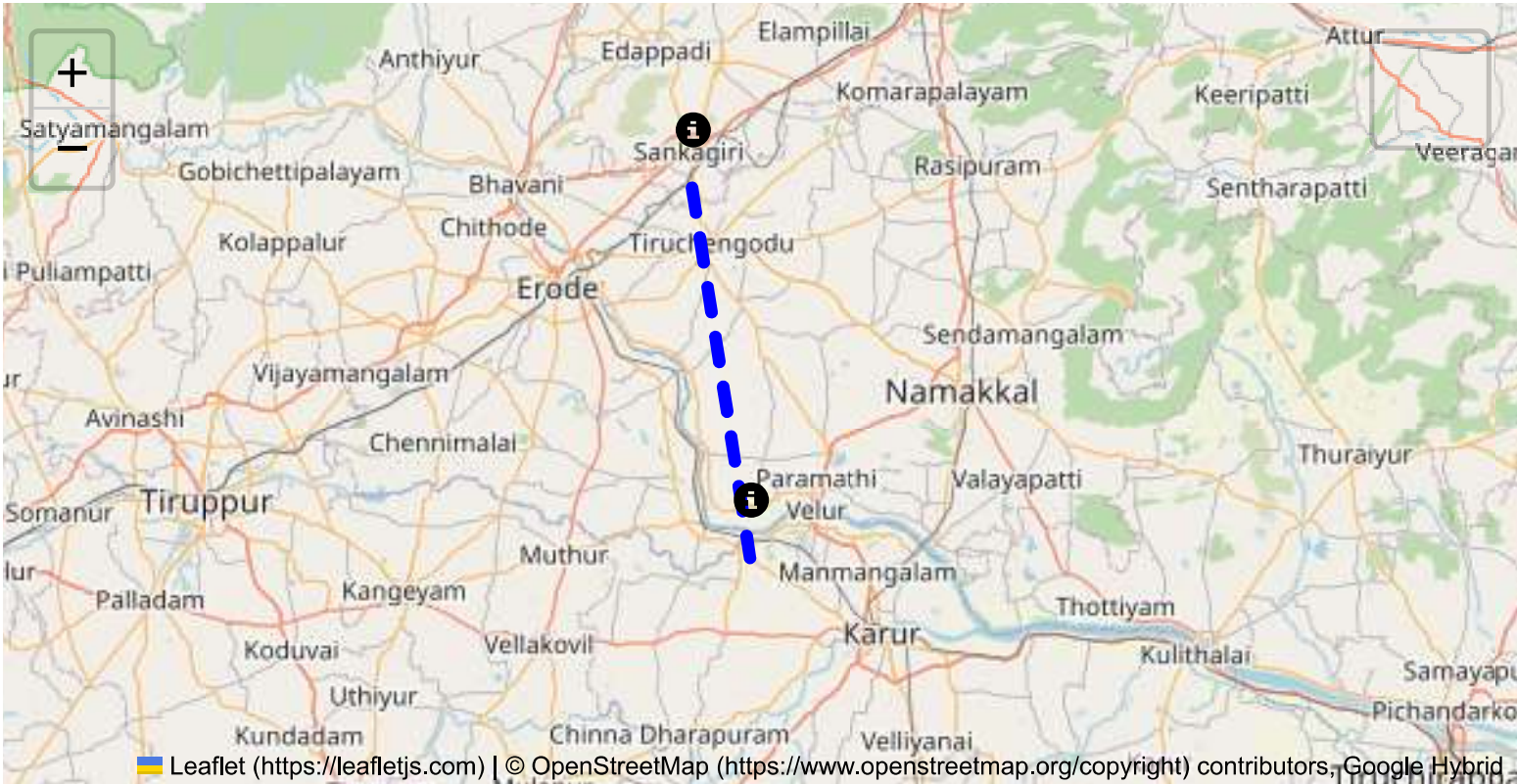
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

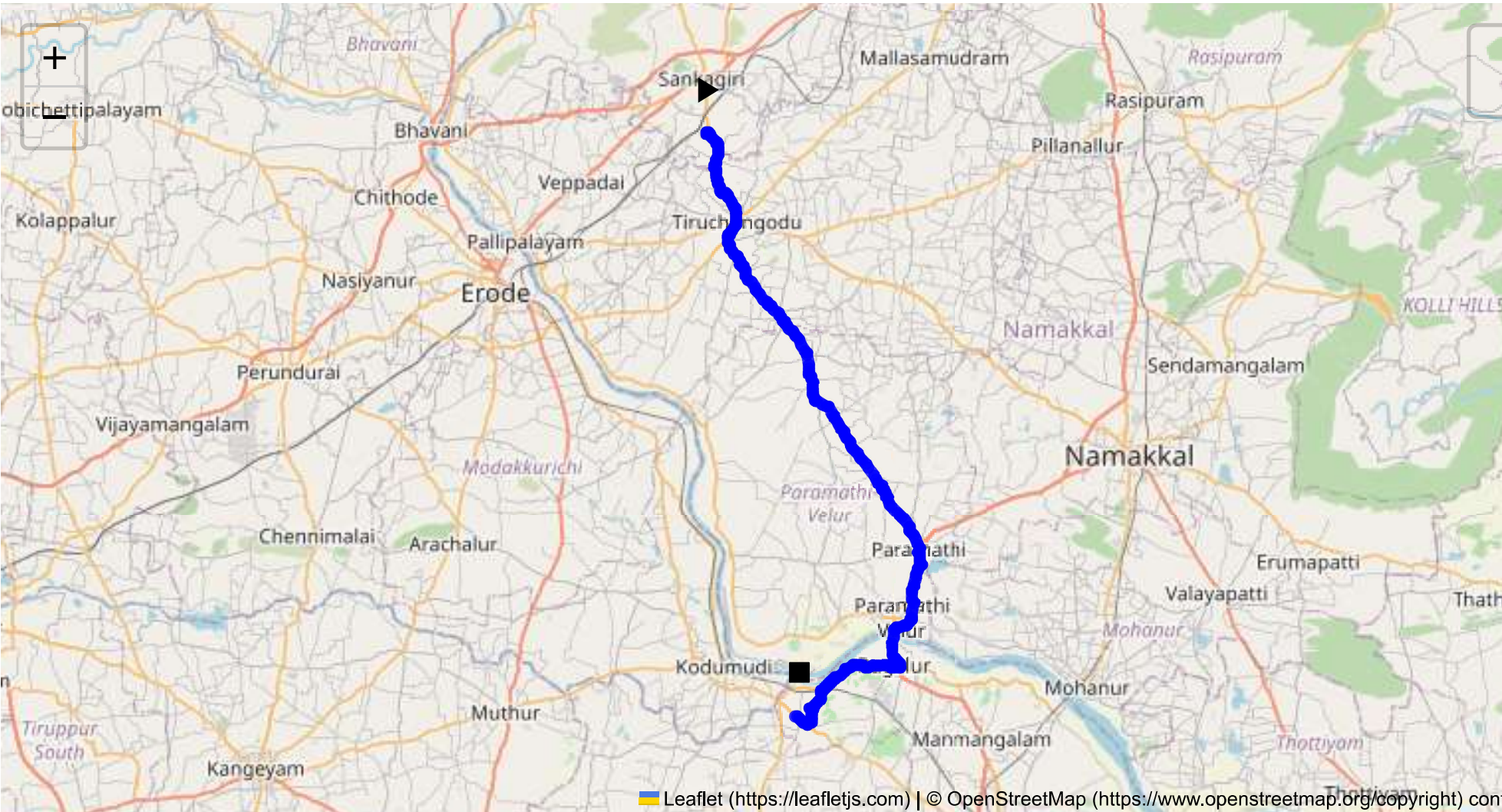
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



**Route Summary:**  
**Total Distance: 59.12 km**  
**Estimated Duration: 1.3 hours**  
**Adjusted Duration (Heavy Vehicle): 1.6 hours**  
**Start: (11.4381, 77.8734)**  
**End: (11.040824, 77.936332)**



## Welcome to the Journey Risk Management Study

### Route Overview

The route from CVQF+23W in Sangagiri to 2WRP+8M3 in Vettamangalam East spans approximately 59.12 km, traversing a combination of rural and semi-urban areas in Tamil Nadu. The journey typically takes around 1.30 hours for heavy vehicles. The main roads likely involved include state highways and district roads, which are common in the region.

# Weather Conditions

Tamil Nadu generally experiences a tropical climate with distinct wet and dry seasons. The southwest monsoon (June to September) and the northeast monsoon (October to December) can bring heavy rainfall and potential flooding, particularly affecting rural and under-maintained roads. During the dry season, high temperatures could potentially lead to dust issues affecting visibility.

# Traffic Patterns

Typical traffic congestion is expected around Sangagiri and in more populated areas along the route. Peak hours usually fall during morning (7-10 AM) and evening (5-8 PM) commutes. Major towns along the route may have localized congestion, especially near schools and markets.

# Road Quality and Infrastructure

The road quality may vary from well-maintained highways to less maintained rural roads. Watch for potholes, unmarked speed bumps, and narrow stretches, especially in less developed areas. Bridges and railway crossings could present additional hazards if not clearly marked or maintained.

# Emergency Routes

For emergencies, alternative routes may involve local service roads connecting through nearby towns like Namakkal or Tiruchengode, depending on the affected stretch. It is essential to have current maps and live traffic updates to recalibrate routes effectively during an emergency.

# Local Regulations

Tamil Nadu has stringent regulations for transporting hazardous materials. These include obtaining necessary permits, adhering to designated routes, and following safety protocols. Ensure proper labeling and handling procedures are observed to comply with local laws.

# Historical Incidents

While detailed incident data is limited, heavy vehicles on rural roads often face risks such as overturning due to narrow roads and sharp bends. The potential for incidents involving hazardous materials would be highest in areas with significant human activity or near water sources.

# Environmental Considerations

Watch for environmentally sensitive areas such as schools, hospitals, and protected natural zones. Avoid stopping or parking near water bodies to prevent contamination risks in case of spills.

# Communication Coverage

While urban areas generally have good mobile coverage, there may be communication dead zones in rural sections. Preloading maps and having offline navigation tools is advisable to mitigate this risk.

# Emergency Response Times

Response times can vary considerably, from under 30 minutes in urban areas to over an hour in more rural sections. Familiarize yourself with the locations of nearby medical facilities, police stations, and fire departments along the route for quicker response in emergencies.

# Overall Risk Assessment

This route presents a moderate risk for transporting hazardous materials due to variable road conditions, weather changes, and traffic congestion, especially around towns. Proper adherence to safety procedures, constant monitoring of weather and traffic reports, and readiness for contingency plans are crucial for minimizing risks. Regular inspection of vehicle condition and ensuring driver familiarity with the route can further bolster safety.

## Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 0  | Roundabout | High       | 11.38395, 77.89480 | 15 KM/Hr    |
| 1  | Roundabout | High       | 11.07462, 78.00608 | 15 KM/Hr    |
| 2  | Roundabout | High       | 11.07485, 78.00156 | 15 KM/Hr    |
| 3  | Turn       | Medium     | 11.43822, 77.87348 | 30 KM/Hr    |
| 4  | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 5  | Blind Spot | Blind Spot | 11.44029, 77.87544 | 10 KM/Hr    |
| 6  | Turn       | High       | 11.37862, 77.89488 | 15 KM/Hr    |
| 7  | Turn       | High       | 11.37850, 77.89453 | 15 KM/Hr    |
| 8  | Turn       | Medium     | 11.16764, 78.01565 | 30 KM/Hr    |
| 9  | Turn       | High       | 11.15573, 78.02176 | 15 KM/Hr    |
| 10 | Turn       | High       | 11.14545, 78.02208 | 15 KM/Hr    |
| 11 | Turn       | High       | 11.14480, 78.02291 | 15 KM/Hr    |
| 12 | Turn       | High       | 11.07467, 78.00842 | 15 KM/Hr    |
| 13 | Turn       | High       | 11.05900, 77.95376 | 15 KM/Hr    |
| 14 | Turn       | High       | 11.04807, 77.94800 | 15 KM/Hr    |
| 15 | Turn       | High       | 11.04661, 77.94755 | 15 KM/Hr    |
| 16 | Turn       | High       | 11.04677, 77.94601 | 15 KM/Hr    |
| 17 | Turn       | High       | 11.04670, 77.94593 | 15 KM/Hr    |
| 18 | Turn       | Medium     | 11.04190, 77.94682 | 30 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 19 | Turn       | Medium     | 11.03753, 77.94595 | 30 KM/Hr    |
| 20 | Blind Spot | Blind Spot | 11.03547, 77.94388 | 10 KM/Hr    |

Emergency Locations

|    | type     | name                                 | coordinates            | speed_limit | risk_level |
|----|----------|--------------------------------------|------------------------|-------------|------------|
| 0  | hospital | Tiruchengode, Goverment Hospital     | 11.3903328, 77.8920627 | 30 km/h     | Medium     |
| 1  | hospital | SPM Medical Centre,Tiruchengode      | 11.3881331, 77.8931963 | 30 km/h     | Medium     |
| 5  | hospital | T.C.A Hospital Tiruchengode          | 11.3791885, 77.8965774 | 30 km/h     | Medium     |
| 6  | clinic   | Kongu Nursing Home                   | 11.3783065, 77.8961134 | 30 km/h     | Medium     |
| 7  | hospital | Soorya Multispecialty Hospital       | 11.3786429, 77.8931912 | 30 km/h     | Medium     |
| 8  | hospital | Tiruchengode Government Hospital     | 11.37645, 77.89426     | 30 km/h     | Medium     |
| 9  | hospital | Krishna Hospital, Namakkal           | 11.3754811, 77.8931817 | 30 km/h     | Medium     |
| 10 | hospital | Tirukumaran Hospitals                | 11.3782118, 77.8914933 | 30 km/h     | Medium     |
| 13 | police   | கரட்டுப்பாளையம் காவல் நிலையம்        | 11.357586, 77.8922539  | 30 km/h     | Medium     |
| 14 | hospital | Goverment Hospital, Nallur           | 11.2679698, 77.946863  | 30 km/h     | Medium     |
| 16 | hospital | Goverment Hospital                   | 11.156647, 78.0213586  | 30 km/h     | Medium     |
| 18 | hospital | Government Hospital, Velauthapalayam | 11.0729521, 78.0023123 | 30 km/h     | Medium     |

Crowded Spots

|    | type        | name                                                           | coordinates               | speed_limit | risk_level |
|----|-------------|----------------------------------------------------------------|---------------------------|-------------|------------|
| 2  | school      | அரசு ஆண்கள் மேல்நிலைப் பள்ளி                                   | 11.3850439,<br>77.8948279 | 30 km/h     | Medium     |
| 3  | school      | அரசு பெண்கள் மேல்நிலைப் பள்ளி                                  | 11.3844247,<br>77.8949053 | 30 km/h     | Medium     |
| 4  | marketplace | திருச்செங்கோடு தினசரி காய்கறி சந்தை                            | 11.3833608,<br>77.8970145 | 30 km/h     | Medium     |
| 11 | school      | KSR Educational institution                                    | 11.3772013,<br>77.8908807 | 30 km/h     | Medium     |
| 12 | school      | MDV School                                                     | 11.3719843,<br>77.8915244 | 30 km/h     | Medium     |
| 15 | school      | Sivabakkiam Muthusamy Matriculation<br>Higher Secondary School | 11.1788177,<br>78.0061696 | 30 km/h     | Medium     |
| 17 | school      | Goverment Higher Secondary School                              | 11.1511724,<br>78.0200723 | 30 km/h     | Medium     |

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.07462, 78.00608



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.07485, 78.00156



**Risk Type: Blind Spot**

**Risk Level: Blind Spot**

**Speed Limit: 10 KM/Hr**

**Coordinates: 11.44029, 77.87544**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.37862, 77.89488**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.37850, 77.89453**



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.16764, 78.01565



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.15573, 78.02176



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.14545, 78.02208



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.14480, 78.02291



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.07467, 78.00842



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.05900, 77.95376**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.04807, 77.94800**



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.04661, 77.94755



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.04677, 77.94601



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.04670, 77.94593



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.04190, 77.94682



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.03753, 77.94595



**Risk Type:** Blind Spot

**Risk Level:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Coordinates:** 11.03547, 77.94388

